CHANDIMA BANDARA

Undergraduate | BSc (Hons) in IT specializing in Data Science

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PROFILE

Highly motivated and detail-oriented undergraduate pursuing a BSc in Information Technology at SLIIT. Equipped with a strong technical foundation in Python, Java, HTML, CSS, and SQL, I am actively seeking a Software Engineering internship or entry level role to apply my hands on coding experience, analytical skills, and problem-solving abilities in building robust, real world software solutions.

EDUCATION

2024 – present Malabe, Sri Lanka	**BSc (Hons) in Information Technology** *Sri Lanka Institute of Information Technology* (*SLIT*)
2022 Dodamgaslandha, Sri Lanka	**Sir John Kotelawala C.C.** I completed my Advanced Level (AL) education with a focus on the technology stream.
2019 Kurunagala, Sri Lanka	**Namal Anga M.V.** I completed my Ordinary Level (OL)

PROJECTS

Student Concern Management System (SCMS)

- **Result:** Developed a role-based concern management platform with end-to-end workflows, email OTP verification, and AI-moderated community forums.
- **Skills & Tools:** Java 21, Spring Boot 4.0.2, Spring MVC, Thymeleaf, Microsoft SQL Server, Spring Data JPA, Spring Security (BCrypt), Google Gemini API, Maven, HTML/CSS
- **Objective:** To build a robust web application for the "Akademy of Knowledge Bridge" that digitizes and manages student concerns, admin workflows, physical meeting bookings, and community engagement.
- Engineered secure user authentication flows including role-based login (Student, Admin, Owner), email OTP verification for registration, and password recovery.
- Developed complex workflow state management for concerns, allowing students to save drafts, upload evidence, submit later, and track resolution status with admin replies.
- Integrated the Google Gemini API to automatically moderate student community posts and replies, ensuring adherence to community rules.
- Built a comprehensive notification center supporting both direct alerts and broadcast messages, complete with a soft-hide feature for student notification cleanup.
- Implemented scheduling logic to facilitate physical meeting proposals between admins and students, including slot booking and decline features.
- GitHub Link

Student Concern Management System (SCMS) – Mobile App

- **Result:** Developed a full-stack, multi-role mobile application to streamline communication between students and academic administration, enabling transparent and efficient resolution of student concerns.
- **Skills & Tools:** React Native, Expo, Node.js, Express.js, MongoDB (Atlas), Mongoose, JWT, Bcryptjs, Multer, Nodemailer, Axios
- **Objective:** To provide a centralized platform where students can submit and track academic, financial, medical, and personal concerns, while administrators manage, triage, and resolve issues with real-time updates and analytics.
- Architected a full MERN stack system with a structured separation of frontend (React Native/Expo) and backend (Node.js/Express), following a clean MVC pattern across controllers, models, and routes.

- Implemented role-based access control supporting three distinct user roles — Student, Admin/Consulter, and Owner — each with dedicated dashboards, permissions, and workflows.
- Built a real-time concern tracking pipeline allowing students to monitor submission status from Pending → Reviewing → Resolved, with in-app and email notifications via Nodemailer.
- Developed a secure medical triage module with file upload support (Multer) and restricted access for Consulters to review sensitive health documentation.
- Integrated executive-level analytics and reporting features, including PDF/Excel report generation, broadcast announcements, and performance metrics for institutional oversight.
- **GitHub Link** 🔗

Semantic Priority Analysis – Campus LMS Support Triage System

- **Result:** Developed an end-to-end NLP and machine learning pipeline to automatically classify student LMS support reports into HIGH, MEDIUM, and LOW priority levels, enabling faster and more efficient triage by campus support staff.
- **Skills & Tools:** Python, Jupyter Notebook, Pandas, Scikit-learn, XGBoost, Transformers (DistilBERT, BART MNLI), NLTK, SMOTE, TF-IDF, Plotly, Seaborn
- **Objective:** To automate the prioritization of student support requests using semantic understanding and machine learning, reducing manual triage delays and ensuring urgent academic or personal concerns reach staff promptly.
- Designed a hybrid labeling strategy combining rule-based keyword dictionaries, zero-shot semantic classification (BART MNLI), and synthetic data augmentation to generate reliable priority labels from a 1005-record dataset.
- Built and compared multiple classification models including Logistic Regression, SVM, Random Forest, XGBoost, Gradient Boosting, and fine-tuned DistilBERT, evaluating each with confusion matrices and visual diagnostics.
- Applied advanced preprocessing techniques including text cleaning, lemmatization, TF-IDF feature extraction, metadata encoding, stratified train/test splitting, and SMOTE-based class balancing to address data imbalance.
- Fine-tuned a DistilBERT transformer model and exported trained artifacts alongside a reusable predict_priority function that returns a predicted class with confidence score for new incoming reports.
- **GitHub Link** 🔗

Student Mental Health Early Detection System

- **Result:** Developed an end-to-end machine learning pipeline to detect early signs of mental health risk in students based on academic and stress-related factors.
- **Skills & Tools:** Python, Jupyter Notebook, Pandas, Scikit-learn, Keras/TensorFlow, Multilayer Perceptron (MLP), Exploratory Data Analysis (EDA), Data Preprocessing
- **Objective:** To identify at-risk students early using academic and self-reported stress data, enabling universities to provide timely support and interventions.
- Led the outlier removal preprocessing phase to clean and structure tabular data from a comprehensive Kaggle dataset.
- Trained and evaluated a Multilayer Perceptron (MLP) neural network, which emerged as the best-performing model with an overall accuracy of 90.91%.
- Designed and executed a group modeling pipeline comparing multiple algorithms including KNN, SVM, Logistic Regression, Decision Trees, and Random Forests.
- Generated comprehensive evaluation metrics, including confusion matrices, accuracy/loss curves, and classification reports to validate model performance.
- GitHub Link 🔗

VeloRent - Web-Based Car Rental System

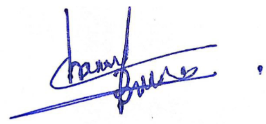
- **Result:** Developed a comprehensive web application to automate the vehicle rental lifecycle with real-time booking and secure authentication.
- **Skills & Tools:** Java 17, Spring Boot, Microsoft SQL Server, HTML, CSS, JavaScript, Thymeleaf, Maven, RESTful APIs, Git
- **Objective:** To build an end-to-end digital platform replacing manual car rental tracking with a scalable, role-based management system.
- Engineered core system security as Lead Developer, implementing user login, registration, Two-Factor Authentication (2FA), and password recovery.
- Developed complex backend logic and user interfaces for the Fleet Manager Dashboard to track real-time vehicle availability, active rentals, and returns.
- Implemented Role-Based Access Control (RBAC) to securely manage distinct permissions across Customers, Owners, Fleet Managers, and Admins.
- Collaborated in a team environment to integrate diverse modules including secure payment processing, vehicle inventory, and dynamic pricing rules.
- GitHub Link 🔗

SKILLS

Python, Java, Spring Boot, MERN, MongoDB, MYSQL,
Machine Learning, Git

DECLARATION

I truly and sincerely state that all above mentioned information are furnished by me and correct to the best of my knowledge.



Chandima Bandara
6th April 2026